The Positive-Chamber Koide Identity for Icosahedral Face Circulants

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Abstract

The stabilizer of an oriented icosahedral face acts regularly on its three corners. Every Hermitian response commuting with this cyclic action is a circulant $C = aI + bR + \bar{b}R^2$, $R^3 = I$. In the chamber where all three eigenvalues are nonnegative and may be read as square-root masses, its Koide invariant is

$$Q = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2.$$

Thus $Q = 2/3$ is exactly equivalent to the single balance condition $|b|/a = 1/\sqrt{2}$. The phase of b drops out of Q and continues to control the two mass ratios jointly. A finite tracial Gelfand–Naimark–Segal construction that pairs the neutral line with a two-state orientation record and the charged plane with one oriented event gives the balanced modulus as a conditional theorem; the balance premise was first abstracted from the measured charged-lepton triple and is algebraically the relation Koide published. Under the balance and mass-ordering premises the measured electron and muon masses fix the tau mass inside a 72-eV enclosure centered at 1776.969027 MeV. The world average lies 0.43 and Belle II 2023 lies 0.89 of their standard uncertainties away; both comparisons are retrospective, and a registered decision rule with a 0.045 MeV precision floor governs later measurements. Koide's relation gives the same mass for every input, so their likelihood ratio is one. The local face fiber has no constructed physical chiral-family attachment, so the result is a representation-theoretic identity, a finite conditional implication and a conditional tau-mass window; no source-only mass prediction is claimed.

Keywords: Koide relation, circulant matrices, cyclic symmetry, icosahedral symmetry, finite event algebra, GNS representation

1 The face-corner carrier

Koide's charged-lepton relation [1] is usually written in terms of the three positive square-root masses. The construction below identifies the exact condition it imposes on a cyclic Hermitian response.

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Let $G = A_5$, the alternating group on five letters, act by proper rotations on the icosahedron. The twenty outward oriented faces form the transitive orbit G/C_3. The stabilizer of one face cyclically permutes its three corners, so every face carries a local copy of the regular C_3 representation. If R denotes the cyclic shift, then

$$R^3 = I, \qquad R^\dagger = R^2.$$

The commutant of this regular action is the three-dimensional circulant algebra

$$\{x_0 I + x_1 R + x_2 R^2 : x_j \in \mathbb{C}\}.$$

Hermiticity restricts a response to

$$C = aI + bR + \bar{b}R^2, \qquad a \in \mathbb{R}, \quad b \in \mathbb{C}. \tag{1}$$

Write $b = \rho e^{i\delta}$, with $\rho = |b| \geq 0$. Fourier diagonalization gives

$$\lambda_k = a + 2\rho \cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2. \tag{2}$$

The unordered spectrum is independent of the chosen face representative. This is a local bundle statement. The sixty face-corner flags form the regular A_5 torsor; the geometry alone supplies no canonical global three-dimensional physical family space.

2 The positive-chamber identity

Theorem 1 (Positive-chamber Koide identity)**.** *Let C be the Hermitian circulant in Eq. (1), with $a > 0$. Suppose its three eigenvalues in Eq. (2) are nonnegative and, for one $s > 0$,*

$$\sqrt{m_k} = \sqrt{s}\lambda_k.$$

Then

$$Q := \frac{\sum_{k=0}^{2} m_k}{\left(\sum_{k=0}^{2} \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{\rho}{a}\right)^2. \tag{3}$$

Consequently,

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad \frac{\rho}{a} = \frac{1}{\sqrt{2}}. \tag{4}$$

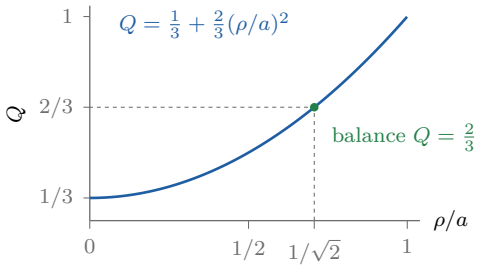
Proof. Set $c_k = \cos(\delta + 2\pi k/3)$. The roots-of-unity identities give

$$\sum_{k=0}^{2} c_k = 0, \qquad \sum_{k=0}^{2} c_k^2 = \frac{3}{2}.$$

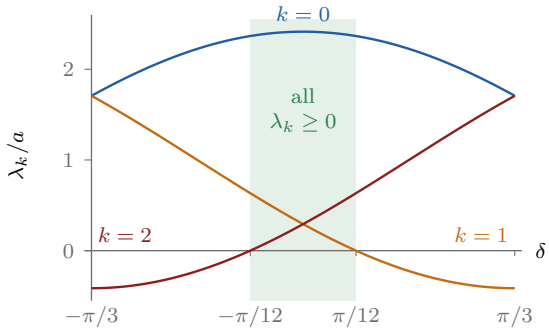
It follows that

$$\sum_k \lambda_k = 3a, \qquad \sum_k \lambda_k^2 = 3a^2 + 6\rho^2.$$

The scale s cancels from Q, yielding Eq. (3). Since $\rho/a \geq 0$, Eq. (3) equals $2/3$ exactly at $\rho/a = 1/\sqrt{2}$. □



(a) Koide ratio when $\sqrt{m_k} \propto \lambda_k \geq 0$



(b) $\lambda_k/a = 1 + \sqrt{2}\cos(\delta + 2\pi k/3)$ at balance; the positive chamber is $|\delta| \leq \pi/12$ modulo $2\pi/3$

Figure 1: (a) The Koide ratio of a nonnegative spectrum of $C = aI + bR + \bar{b}R^2$, $\rho = |b|$, as a function of ρ/a (Theorem 1); balance $\rho/a = 1/\sqrt{2}$ gives $Q = 2/3$. (b) At balance the three eigenvalues are nonnegative exactly in the chamber $|\delta| \leq \pi/12$ modulo $2\pi/3$, the chamber outside which the theorem makes no statement.

Remark 2 (The chamber is part of the theorem). The signed trace calculation holds algebraically for every δ. Physical square roots use $|\lambda_k|$ when an eigenvalue is negative, and then the denominator is not the signed trace. At $\rho/a = 1/\sqrt{2}$, the positive chamber is

$$|\delta| \leq \frac{\pi}{12} \pmod{\frac{2\pi}{3}}.$$

Theorem 1 makes no physical Koide statement outside that chamber.

Remark 3 (What the phase contains). Inside the positive chamber, Q is independent of δ. The three eigenvalues, and hence the two reported mass ratios, vary with δ. The balance condition removes one scale-free degree of freedom from a three-mass spectrum. It does not determine the ratios.

3 Conditional finite tracial balance

The balance condition in Eq. (4) can arise exactly from a finite event packet. This subsection states the packet as an independent conditional theorem. Its premise ancestry comes first: the balance premise was first abstracted from the measured charged-lepton triple, and it is algebraically the relation Koide published for those masses. The packet below realizes it through an asymmetric block choice, which is where that premise enters.

Let

$$\mathcal{V} = \mathbf{1} \oplus \chi \oplus \overline{\chi}, \qquad \mathcal{H}_{\mathrm{or}} = \mathbb{C}^2, \qquad \mathcal{A} = B(\mathcal{V} \otimes \mathcal{H}_{\mathrm{or}}) \simeq M_6(\mathbb{C}).$$

Here $\mathbf{1}$ is the neutral cyclic mode, $\chi \oplus \overline{\chi}$ is the two-dimensional charged plane, and $\mathcal{H}_{\mathrm{or}}$ is a two-state orientation record. Let P_0 and P_c project onto the neutral line and charged plane. For a rank-one oriented event q_+, define

$$Z_0 = P_0 \otimes I_{\mathrm{or}}, \qquad Z_c = P_c \otimes q_+, \qquad E_+ = Z_0 + Z_c.$$

Both Z_0 and Z_c have rank two.

Theorem 4 (Conditional finite tracial-GNS balance)**.** *Condition the normalized trace state $I_6/6$ on E_+. Map the orthonormal cyclic basis of the resulting square-root amplitude to (I, R, R^2) in $L^2(M_3(\mathbb{C}), \tau_3)$. Then*

$$p_0 = p_c = \frac{1}{2}, \qquad a = \sqrt{p_0}, \qquad \rho = \sqrt{\frac{p_c}{2}},$$

and therefore

$$\frac{\rho}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}$$

whenever the positive square-root-mass reading of Theorem 1 is supplied.

Proof. Normalized-trace conditioning assigns probability proportional to block rank. Both accepted blocks have rank two, so $p_0 = p_c = 1/2$. In $L^2(M_3(\mathbb{C}), \tau_3)$, the operators I, R, R^2 are orthonormal. The canonical square-root vector therefore has neutral coefficient $\sqrt{p_0}$ and two charged coefficients of modulus $\sqrt{p_c/2}$. Their ratio is $1/\sqrt{2}$. Theorem 1 then gives $Q = 2/3$ on the positive chamber. □

Theorem 4 is finite and exact. Its event algebra, block choice, conditioned trace, and cyclic response map are premises. The block choice is asymmetric: the neutral line is paired with the full two-state orientation record and the charged plane with one oriented event. Treating the two blocks alike, both with I_{or} or both with q_+, gives $p_c/p_0 = 2$, hence $\rho/a = 1$ and $Q = 1$. The normalized trace fixes the probabilities only after the asymmetric event has been chosen, so the balanced modulus is the content of that choice. A physical charged-lepton statement additionally requires a quotient-visible map from the local face bundle to one chiral three-family response, preservation of the relevant block powers, and a mass readout. Those objects are absent from the theorem. The executable observer construction supplies none of them: the tau window consumes no native observer run, and its identification of balanced square-root masses with charged-lepton pole masses is a premise [7, 6].

4 The conditional tau-mass window

Fixing $Q = 2/3$ in Eq. (3) and inserting the measured electron and muon masses turns the Koide coordinate into one quadratic in $\sqrt{m_\tau}$. The mass-ordering premise $m_\tau > m_\mu$ selects the physical root; the excluded second root sits at 3.317 MeV, below the muon mass. Propagating the one-sigma corners of the measured (m_e, m_μ) inputs [2] with outward rounding encloses the selected root in

$$m_\tau \in [1776.968991,\ 1776.969063]\ \mathrm{MeV}, \tag{5}$$

a 72-eV window with center 1776.969027 MeV. The center sits 0.43 standard uncertainties from the measured 1776.93 ± 0.09 MeV, and the window is three orders of magnitude narrower than that measurement uncertainty. The dedicated Belle II 2023 measurement, $1777.09 \pm 0.08_{\mathrm{stat}} \pm 0.11_{\mathrm{syst}}$ MeV, lies 0.12097 MeV above the center, 0.89 of its combined 0.136 MeV uncertainty [4]. That comparison is retrospective, and its systematic uncertainty alone exceeds the 0.045 MeV compatibility floor, so it carries no verdict under the registered rule [5].

The comparison is governed by a registered conditional test whose target definition and decision rule were fixed in a frozen prediction record [3]: a world-average or dedicated-measurement central value more than three standard uncertainties from 1776.969027 MeV refutes the balanced-circulant premise; compatibility requires agreement within two standard uncertainties at a measurement uncertainty of at most 0.045 MeV; every other outcome carries no verdict. The premise ancestry

is declared: the balance premise, the equal-weight condition on the two response sectors of one icosahedral face circulant, was first abstracted from the measured charged-lepton triple, and it is algebraically the relation Koide published for those masses. The window is therefore a conditional postdiction whose evidential weight lies in the kill direction [3, 6]. Improving tau-mass measurements test the balanced-circulant premise and Koide's relation together. Both send every allowed (m_e, m_μ) to the same m_τ, so under one measurement model their likelihood ratio is one and no precision in this observable distinguishes them; the formula contains no screen-grain coordinate P and constrains no other OPH parameter [6].

5 Numerical diagnostic and provenance

Using the Particle Data Group 2026 central masses [2],

$$(m_e, m_\mu, m_\tau) = (0.51099895069,\ 105.6583755,\ 1776.93)\ \text{MeV},$$

gives

$$Q_{\text{PDG}} = 0.6666644634026367.$$

The declared minimal complete public response model gives

$$\begin{aligned} Q_{\text{MCPR}} &= 0.6666644634090389, \\ \rho/a &= 0.7071044442750720, \\ \frac{\rho/a - 1/\sqrt{2}}{1/\sqrt{2}} &= -3.3049 \times 10^{-6}. \end{aligned}$$

The response dimensions, path table, amplitude, phase, and determinant exponent are target-informed inputs. The numerical response is therefore neither blind nor source-derived. The proximity of the two displayed Q values carries no significance or predictive weight.

Statement	Scope
Eq. (3)	exact algebra in the nonnegative-eigenvalue chamber
Theorem 4	exact implication conditional on the declared finite event packet
Icosahedral face fiber	exact local A_5/C_3 bundle and unordered spectrum
Tau window, Eq. (5)	exact quadratic consequence of the balance and ordering premises; registered decision rule
Physical family attachment	absent from the theorems; a premise of any physical reading
Phase and numerical ratios	balance leaves one phase controlling both scale-free ratios
MCPR numerical proximity	target-informed comparison-only diagnostic

6 Why residual symmetry does not finish the ratios

The family-shape problem can be stated in the real five-dimensional representation

$$W_5 \simeq \text{Sym}^2_0(\mathbb{R}^3),$$

the traceless symmetric 3×3 matrices with action $A \mapsto gAg^T$.

Proposition 5 (Residual-stabilizer boundary)**.** *Invariance under a threefold or fivefold rotation about an axis n forces*

$$A = \alpha(nn^T - I/3),$$

which has a double eigenvalue. Invariance under a twofold rotation has a three-dimensional linear fixed locus, hence two projective parameters, and that locus admits simple spectrum.

Proof. Choose the rotation axis as the third coordinate. Decompose a traceless symmetric matrix into a planar scalar, planar spin-two anisotropy, transverse vector, and axial entry. Rotation by θ acts on the anisotropy by 2θ and on the transverse vector by θ. For $\theta = 2\pi/3$ or $2\pi/5$, neither block has a nonzero fixed vector, so only the axial combination remains. For $\theta = \pi$, the planar symmetric block survives:

$$A = \begin{pmatrix} u & v & 0 \\ v & w & 0 \\ 0 & 0 & -u-w \end{pmatrix}.$$

This space has dimension three and contains matrices with three distinct eigenvalues. □

Threefold and fivefold symmetry cannot produce three distinct masses. Twofold symmetry leaves two scale-free coordinates, exactly enough to carry the two mass ratios without selecting them. A numerical spectrum therefore requires a specific invariant potential derived from the screen dynamics. This proposition locates the missing input; it supplies no replacement fit.

7 Formal proof boundary

The algebraic core has a Lean 4 formalization. It proves the root sum, square sum, quotient formula, reciprocal-square-root identity, and the nonnegative-modulus equivalence

$$\frac{1}{3} + \frac{2}{3}r^2 = \frac{2}{3} \quad \Longleftrightarrow \quad r = \frac{1}{\sqrt{2}}.$$

The formal module deliberately does not identify signed eigenvalues with physical square roots. Positivity of all three eigenvalues, the phase selection, and the source-to-charged-family attachment are outside its statement. This division matches Theorem 1: the checked algebra is exact, and the physical reading is a separate premise.

8 Conclusion

The regular three-corner face carrier turns the Koide relation into one transparent modulus condition. In the positive chamber,

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad \rho/a = \frac{1}{\sqrt{2}}.$$

The finite tracial event packet supplies this balance conditionally through equal block weights and the canonical square-root representation. Under the balance and mass-ordering premises the measured electron and muon masses fix the tau mass inside the 72-eV window of Eq. (5), and a

registered decision rule fixes which later measurements refute the balanced-circulant premise; the world-average and Belle II comparisons are retrospective. The result does not determine the phase or mass ratios and does not attach the local face fiber to physical charged leptons. Its durable content is an exact circulant identity, an exact finite conditional balance theorem, a conditional tau-mass window with a registered decision rule, and a sharp boundary between the checked algebra and the physical attachment premises.

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